📝 Worksheet W12: Police Proofs

"Challenge Problems — Prove You Understand the Calculus Code"

Epilogue · 10 Problems · Name: ______________________ Date: ______________________

💡 A Police Proof requires reasoning and explanation — not just calculation. Write your answer as a complete explanation.

Problem P1

Question: Why is calculus called "arithmetic for things that move"? Explain using the concepts of **derivative** and **integral**.

💡 *Hint: Think about the speedometer and odometer.*

Problem P2

Question: Explain the **0/0 paradox** and how the **limit** trick solves it.

💡 *Hint: Speed = Distance ÷ Time. What are the values at an instant?*

Problem P3

Question: Explain why **slope** and **speed** are the same concept. Use the ideas of **rise over run** and the **Mental Zoom**.

Hint: Both are "one change divided by another."

Problem P4

Question: In your own words, explain the **Power Rule**. Why is it a "shortcut," and how does it relate to the zoom-in method?

Hint: What does the Power Rule do? How long does it take compared to the zoom-in method?

Problem P5

Question: Explain the relationship between the **derivative** and the **integral**. Use the analogy of a **speedometer** and **odometer**.

Hint: The derivative and integral are opposites — they undo each other.

Problem P6

Question: Explain why e^x is called the **"super-function."** What makes it special compared to other functions?

💡 *Hint: What is the derivative of e^x? What is the integral of e^x?*

Problem P7

Question: Explain the **derivative cycle** for sine and cosine. Why does it take four steps to return to the start?

💡 *Hint: sin → cos → -sin → -cos → back to sin.*

Problem P8

Question: Explain why the **derivative is zero** at both maximums and minimums. Use the hill and valley analogy.

💡 *Hint: At the top of a hill, the slope is zero. At the bottom of a valley, the slope is also zero.*

Problem P9

Question: Explain how the **Mental Zoom** connects **slope**, **speed**, **area**, and **accumulation**.

💡 *Hint: What happens when you zoom in on a curve? What happens when you zoom out and add more rectangles?*

Problem P10

Question: Why is the **Fundamental Theorem of Calculus** called the "Great Reversal"? Explain its importance in your own words.

💡 *Hint: Before the theorem, slope (derivative) and area (integral) were seen as separate. What did the theorem reveal?*